

# ENGR 200

## Probability and Random Variables for Engineers

Introduction

# Course Outline

- ❑ Introduction to Probability
  - ❑ Where to use probability theory
- ❑ Probabilistic Models
  - ❑ How to assign probabilities to outcomes
- ❑ Discrete Random Variables
  - ❑ How to assign probabilities to numerical outcomes
- ❑ Continuous Random Variables
  - ❑ How to deal with an infinite number of outcomes
- ❑ Transforms of Random Variables
  - ❑ How to add, transform random variables
- ❑ Limit Theorems
  - ❑ Probability as relative frequency in repeated trials
- ❑ Random Process
  - ❑ How to assign probabilities to functions evolving in time

# Course Outline

- ❑ Introduction to Probability

- ❑ Where to use probability theory

- ❑ Probabilistic Models

- ❑ How to assign probabilities to outcomes

- ❑ Discrete Random Variables

- ❑ How to assign probabilities to numerical outcomes

- ❑ Continuous Random Variables

- ❑ How to deal with an infinite number of outcomes

- ❑ Transforms of Random Variables

- ❑ How to add, transform random variables

- ❑ Limit Theorems

- ❑ Probability as relative frequency in repeated trials

- ❑ Random Process (as time allows)

- ❑ How to assign probabilities to functions evolving in time

# Where to Use Probability?

- In your daily life
  - Use words such as probable/improbable, likely/unlikely
- In many fields of electrical and computer engineering
  - Communications
  - Networking
  - Control theory
  - Statistical signal processing

# Where to Use Probability?

- In your daily life
  - Use words such as probable/improbable, likely/unlikely
- In many fields of electrical and computer engineering
  - Communications
  - Networking
  - Control theory
  - Statistical signal processing

# In Daily Life

*Dialog from your book: A patient admitted to the hospital and a potentially life-saving drug is administered. Here is the dialog between the nurse and a concerned relative:*

- R: Nurse, what is the probability that the drug will work?
- N: I hope it works, we'll know tomorrow
- R: Yes, but what is the probability that it will?
- N: Each case is different, we have to wait.
- R: But let's see, out of a hundred patients that are treated under similar conditions, how many times would you expect it to work?
- N: I told you, every person is different, for some it works, for some it doesn't.
- R: Then tell me, if you had to bet whether it will work or not, which side of the bet would you take?
- N: I'd bet it would work.

# In Daily Life

- *Relative* use the concept of probability to discuss an ***uncertain*** situation
- *Statement 1*: Out of a hundred patients that are treated under similar conditions, how many times would you expect it to work?
  - **Frequency of occurrence**: percentage of successes in a moderately large number of similar situations
  - Nurse may be right: Maybe she did NOT experience many similar situations

# In Daily Life

- *Statement 2:* If you had to bet whether it will work or not, which side of the bet would you take?
  - **Subjective belief:** using person's past experiences in other situations to judge another case
  - "I'd bet it would work.": The probability of success is at least 50%.

# In Daily Life

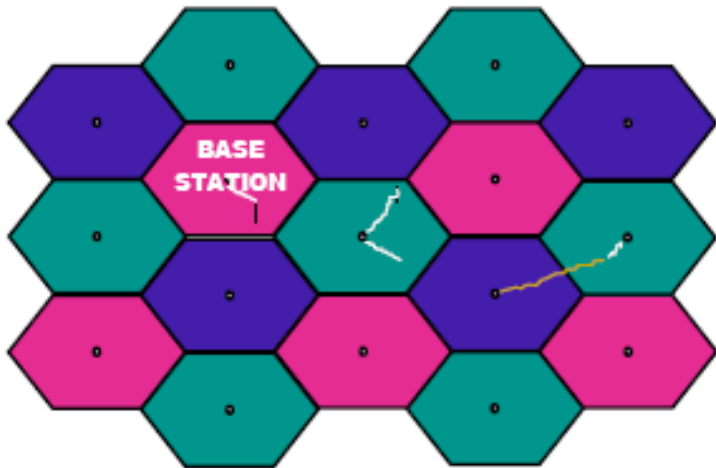
- ***Probabilistic notions are commonplace in everyday language***
- We use words such as
  - Probable/improbable, possible/impossible
  - Certain/uncertain, likely/unlikely
- We use phrases such as
  - There is a 50-50 chance
  - The probability of precipitation is 20%

# Where to Use Probability?

- In your daily life
  - Use words such as probable/improbable, likely/unlikely
- In many fields of electrical and computer engineering
  - Communications
  - Networking
  - Control theory
  - Statistical signal processing

# Communication Systems

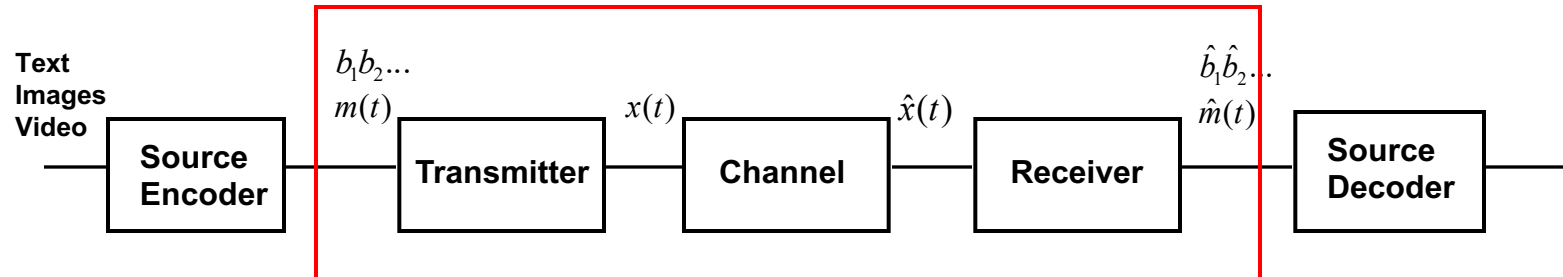
Cellular Systems



Inter-vehicular Communications



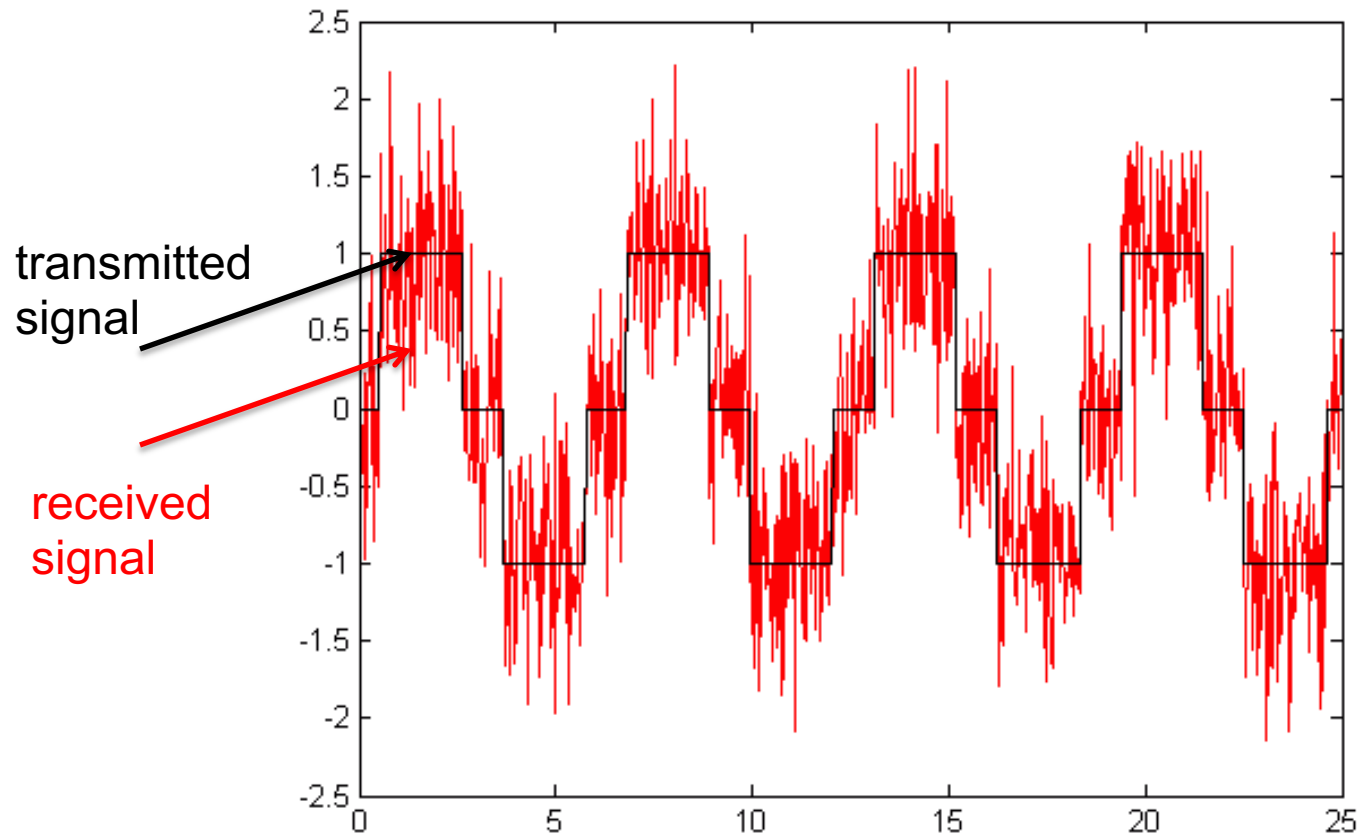
# Communication Systems



- Source encoder converts message into bits
- Transmitter converts message signal or bits into format appropriate for channel transmission
- Channel introduces ***distortion, noise and interference.***
- Receiver decodes received signal back to message signal
- Source decoder decodes message signal back into original signal

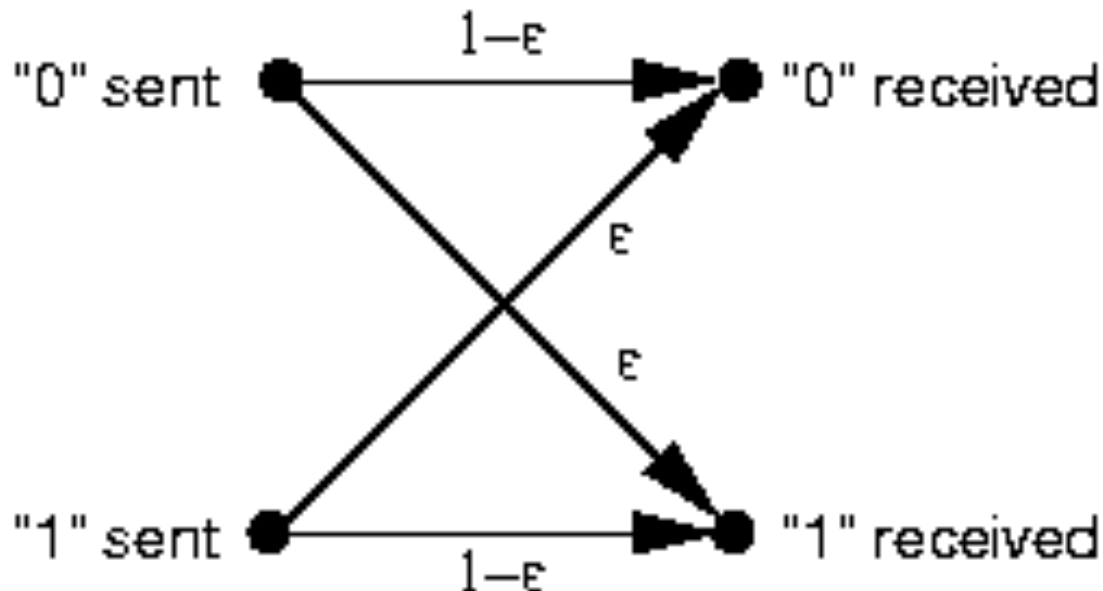
**Goal: Design transmitter and receiver to mitigate distortion and noise from the channel.**

# Channel Distortion



# Binary Symmetric Channel

- A model for a binary communication channel
  - Every  $T$  seconds, the transmitter accepts a binary input, a 0 or a 1, and transmits the corresponding signal
  - At the end of  $T$  seconds, the receiver makes a decision about what the input was
  - Goal: design transmitter and receiver to minimize  $\epsilon$



$\epsilon$  is the long-term proportion of bits delivered in error

# Binary Symmetric Channel

- How to improve performance?
  - Error correcting codes:
    - Simplest example, repetition code:  $0 \rightarrow 000$ ,  $1 \rightarrow 111$
  - Receiver makes a decision by taking a majority vote of 3 bits
    - $0 \rightarrow 000 \rightarrow \text{channel} \rightarrow 000 \rightarrow \text{decide for } 0$
    - $0 \rightarrow 000 \rightarrow \text{channel} \rightarrow 001 \rightarrow \text{decide for } 0$
    - $0 \rightarrow 000 \rightarrow \text{channel} \rightarrow 011 \rightarrow \text{decide for } 1: \text{error}$
    - $0 \rightarrow 000 \rightarrow \text{channel} \rightarrow 111 \rightarrow \text{decide for } 1: \text{error}$
  - What is the probability of error for the coded case?

$$3\varepsilon^2 - 2\varepsilon^3$$

# Repetition Code

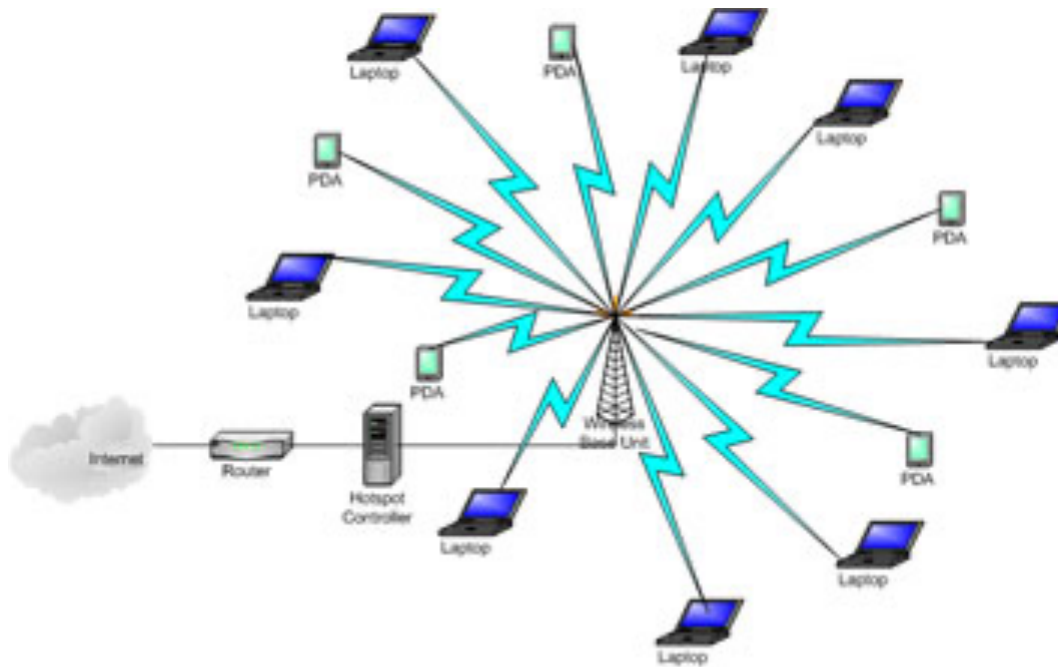
- *The probability of error is decreasing compared to the case without coding*
- The **rate of transmission** has slowed down to 1 bit every  $3T$  seconds

# Where to Use Probability?

- In your daily life
  - Use words such as probable/improbable, likely/unlikely
- In many fields of electrical and computer engineering
  - Communications
  - **Networking**
  - Control theory
  - Statistical signal processing

# Computer Networks

- How to share common medium?
  - Collision if at least two nodes send at the same time

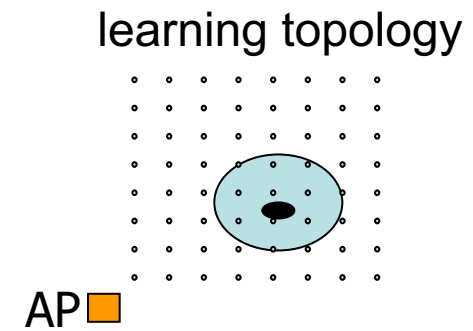
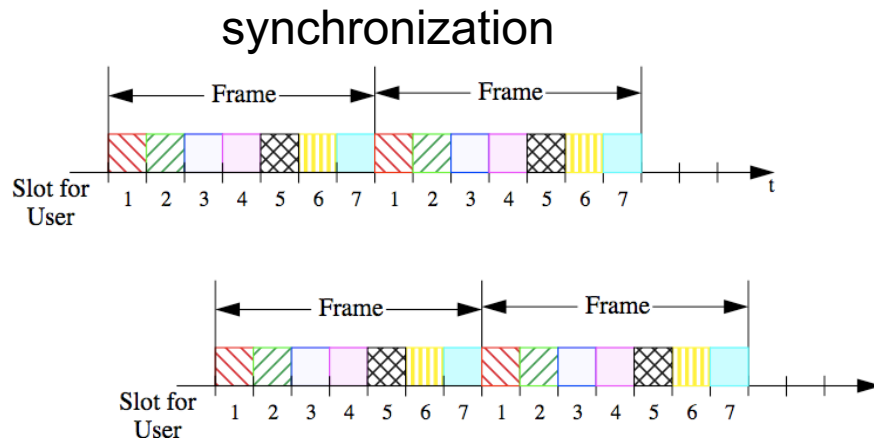


Wireless Star Topology  
Configuration

# Schedule vs. Contention-Based Protocols

- **Schedule-based MAC**

- A **schedule** exists, regulating which participant may use which resource at which time
- Advantages: collision problem solved
- Disadvantages: Time synchronization needed, overhead of learning the topology.



# Schedule vs. Contention-Based Protocols

- ***Contention-based*** protocols
  - Risk of colliding packets is deliberately taken
  - Hope: coordination overhead can be saved, resulting in overall improved efficiency
  - ***Mechanisms to handle/reduce probability/impact of collisions required***

# Where to Use Probability?

- In your daily life
  - Use words such as probable/improbable, likely/unlikely
- In many fields of electrical and computer engineering
  - Communications
  - Networking
  - Control theory
  - Statistical signal processing

# Course Outline

- ❑ Introduction to Probability
  - ❑ Where to use probability theory
- ❑ Probabilistic Models: NEXT CLASS
  - ❑ How to assign probabilities to outcomes
- ❑ Discrete Random Variables
  - ❑ How to assign probabilities to numerical outcomes
- ❑ Continuous Random Variables
  - ❑ How to deal with an infinite number of outcomes
- ❑ Transforms of Random Variables
  - ❑ How to add, transform random variables
- ❑ Limit Theorems
  - ❑ Probability as relative frequency in repeated trials
- ❑ Random Process (as time allows)
  - ❑ How to assign probabilities to functions evolving in time